VA DIVISION A • 2025-2026

Bridge Building

Official Study Guide • 2025–2026 Season

- Official Rules
- Scoring Formula
- Truss Types
- Tension vs Compression
- Build Strategy
- Competition Tips

Event Overview

Rules

Build the TALLEST truss bridge from straws and tape to span a 30 cm gap — hold a 300g or 500g mass for 30 seconds. Score = Height + Assessment. Max = 100 points.

Materials:	30 paper straws (~20 cm each), 1 meter of 2.54 cm masking tape. Nothing else.
Span:	Bridge must span exactly 30.0 cm between the two provided support structures.
Min Width:	Bridge must be at least 10.0 cm wide at center (outside edge to outside edge).
Loading Block:	5×5×0.6 cm block (20g) placed on the top chord at center before the test mass.
Masses:	Hold 300g for 30s for base score. Hold 500g for 30s for 1.2× height bonus.
Clear Path:	No internal cross-members that would block the 5×5 cm block from passing through.
Safety:	ANSI Z87+ impact-rated glasses required. Bring your own — not provided at event.

Official Scoring Formula

Scoring

Team Score = 0.5 × (THS ÷ CHS) + 0.5 × (TAS ÷ CAS) × multipliers • Maximum = 1.0 (normalized)

Term	Meaning	Source/Impact
THS — Team Height Score	Your bridge's measured center height in mm	From measurement
CHS — Competition Height Score	Tallest bridge among all competing teams	Divides THS
TAS — Team Assessment Score	Your written test score	From assessment
CAS — Competition Assessment Score	Highest assessment score among all teams	Divides TAS
Bonus ×1.2	500g mass held for 30s → height score × 1.2	Best outcome
Penalty ×0.5	Width < 10 cm OR loading block can't pass through	Design fault
Height = 0	Bridge fails to span the 30 cm gap	Zero score
Tiebreaker 1	Most unused complete straws remaining	Save material!
Tiebreaker 2	Time to failure in 500g test (longer = better)	Secondary

Example: Your height 320mm, best=400mm → THS/CHS=0.80. Your test 8/10, best=10 → TAS/CAS=0.80. Score = 0.5×0.80 + 0.5×0.80 = 0.80. With 500g bonus: 0.40 +

Truss Bridge Components

Compone
nts

A truss bridge carries loads through a framework of triangles. Every component has a specific role — and knowing whether each member carries tension or compression is tested on the written assessment.

TOP CHORD

Upper horizontal member running the bridge's full length.

Force: Usually in COMPRESSION under a center load.

★ Most likely to buckle — bundle straws or slide one inside another for strength.

BOTTOM CHORD

Lower horizontal member at deck level.

Force: Usually in TENSION — being pulled apart.

★ Tape joints well — tension failures happen at connections, not mid-member.

VERTICALS

Members perpendicular to the chords.

Force: Compression (Howe/Warren) or Tension (Pratt) depending on truss type.

★ In Pratt truss: verticals carry compression. In Howe: verticals carry tension.

DIAGONALS

Angled members connecting top and bottom chords.

Force: Pratt = tension. Howe = compression. Warren = alternates.

★ Direction of diagonals defines the truss type — memorize the difference for the test!

JOINTS / NODES

Connection points where members meet.

Force: Forces assumed to act only at joints in ideal truss analysis.

LOADING BLOCK AREA

5×5 cm block sits on top chord at center.

Force: Concentrated load applies maximum bending moment at bridge center.

Types of Truss Bridges

Truss Types

PRATT TRUSS

Diagonals → tension

Verticals → compression

Most common design. Diagonals slope TOWARD the center.

★ Diagonal direction: lean inward, toward center.

HOWE TRUSS

Diagonals → compression

Verticals → tension

Opposite of Pratt. Diagonals slope AWAY from the center.

★ Diagonal direction: lean outward, away from center.

WARREN TRUSS

Alternates tension/compression

No verticals (basic form)

Equilateral triangles with alternating diagonal directions. Simple and efficient.

★ No verticals in basic Warren — only alternating diagonals.

PARKER TRUSS

Like Pratt (tension)

Like Pratt (compression)

Pratt variant with a curved top chord — deeper at center where moments are highest.

★ Also called 'camelback truss' for its arched profile.

BOWSTRING TRUSS

N/A — arched top chord

Varies

Arched top chord (in compression like a bow) with straight bottom chord in tension.

★ Bottom chord acts like a bowstring. The arch carries loads by compression.

K-TRUSS

Two diagonals meet each vertical at mid-height

Central vertical

Named for K-shaped panels. Used for tall long-span bridges.

★ Each panel looks like the letter K — one diagonal up, one diagonal down.

⚡ Tension, Compression & Force Analysis

Forces

TENSION

- ◆ Forces PULL the member apart (stretching)
- ◆ Member elongates under tension
- ◆ Bottom chord of a loaded bridge = tension
- ◆ Diagonal members in Pratt truss = tension
- ◆ Straw bridges fail in tension at joints — tape carefully!
- ◆ Tension members can be made from thinner material
- ◆ String, wire, and rubber bands resist tension well

COMPRESSION

- ◆ Forces PUSH the member together (squeezing)
- ◆ Member shortens or BUCKLES under compression
- ◆ Top chord of a loaded bridge = compression
- ◆ Diagonal members in Howe truss = compression
- ◆ Straw bridges fail in compression by buckling!
- ◆ Bundle straws or slide one inside another to resist
- ◆ Short, fat columns resist buckling better than long thin ones

ANALYSIS METHODS

- ◆ Method of Joints: apply $\Sigma F_x=0$ and $\Sigma F_y=0$ at each joint
- ◆ Start at a joint with only 2 unknown forces
- ◆ Unknown = member force (+ tension, – compression)
- ◆ Method of Sections: cut through 3 members max
- ◆ Take equilibrium of one half to find cut member forces
- ◆ Useful when only a few members need analysis
- ◆ Both methods assume pin (hinge) joints — no bending at joints

🔧 Build Strategy — 25-Minute Session

Strategy

1 Design first (0–3 min)

Sketch your Warren or Pratt truss. Count straws: ~10 for bottom chord + verticals, ~8 for top chord, ~8 for diagonals, 4 reserve. Set target height (aim high!).

2 Assign roles

One student takes the written test. The other leads the build. Both must know the 10cm width requirement and know not to block the 5×5 loading block path.

3 Build one truss side first (3–12 min)

Lay straws flat to form one side of the truss. Connect diagonals before lifting. Let tape set by pressing each joint firmly. Check width is ≥10 cm.

4 Build second side + cross-bracing (12–20 min)

Mirror the first side. Add cross-bracing connecting both sides — X-patterns on the faces resist racking. Cross-bracing also increases height if done vertically.

5 Reinforce and measure (20–24 min)

Measure center height to the nearest mm. Reinforce any loose joints. Check loading block path is clear. Measure width one final time — must be ≥10 cm!

6 Save your straws (ongoing)

Unused complete straws are Tiebreaker 1. Do NOT cut straws unless necessary. Every whole straw saved is a potential tiebreaker win.

📖 Key Vocabulary & Famous Bridges

Vocab

Truss:	Framework of interconnected triangles that forms the load-bearing structure of a bridge
Chord:	The top or bottom horizontal member running the length of a truss
Diagonal:	Angled member connecting the top and bottom chords
Vertical:	Perpendicular member connecting the top and bottom chords
Tension:	Force that pulls a member apart — member elongates
Compression:	Force that squeezes a member together — member shortens or buckles
Buckling:	When a slender compression member suddenly bends sideways and fails
Method of Joints:	Analyzes truss forces using $\Sigma F_x=0$, $\Sigma F_y=0$ at each joint

Through Truss:	Load passes through the truss (between top and bottom chord)
Deck Truss:	Load travels on top of the truss (on the top chord)
Pratt Truss:	Diagonals in tension, verticals in compression. Diagonals lean inward.
Howe Truss:	Diagonals in compression, verticals in tension. Opposite of Pratt.
Warren Truss:	Alternating diagonals, no verticals in basic form
Bowstring Truss:	Arched top chord (compression) with straight tension bottom chord
THS:	Team Height Score — your bridge's center height in mm
CHS:	Competition Height Score — tallest bridge among all teams